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RESEARCH ARTICLE

Can News Sentiment Improve Deep Learning Models for Nifty 50 Index Forecasting?

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ABSTRACT A stock index, such as the Nifty 50, offers diversified exposure and reduces the risk of investing in individual companies. Index price movements are influenced by internal and external factors, including political, economic, and environmental developments, as well as historical trends. The relationship between news sentiment and the Nifty50 return has not been thoroughly studied. This study examines whether financial news sentiment affects index movements and how sentiment can enhance the prediction of next-day returns. Polarity and subjectivity are extracted from financial news using pre-trained transformer models. Deep learning models, including LSTM, GRU, SimpleRNN, and temporal Kolmogorov-Arnold network (TKAN), are trained on return sign, polarity, and subjectivity using a five-day rolling window to forecast the next-day index return sign. Experimental results demonstrate that the proposed approach outperforms baseline methods, achieving a 5.2% improvement in average accuracy. Incorporating polarity and historical return signs enhances performance across all models. By employing a focused feature set, domain-specific sentiment analysis, and a streamlined architecture, the model achieves superior predictive accuracy. Causal analysis and Shapley Additive Explanations (SHAP) reveal that polarity exhibits a causal effect on returns, while subjectivity does not. The study has practical significance, offering day traders and short-term investors timely, data-driven insights to manage risk and make informed investment choices.

INDEX TERMS Causal analysis, deep learning, FinBERT, GRU, LSTM, particle swarm optimization (PSO), sentiment analysis, temporal TKAN.

I. INTRODUCTION

Stock index movement prediction is challenging for financial experts, investors, and traders. A stock index, such as the Nifty 50, comprises 50 companies across 12 sectors, and major news concerning these companies can influence the overall index movement. Research has shown a causal relationship between sentiment in financial news and stock price changes [1], [2], [3], suggesting that incorporating news sentiment may enhance the accuracy of stock index prediction models. Conducting a causal analysis between financial news sentiment and index returns can help quantify the importance of sentiment scores in forecasting stock movements [4].

Natural language processing (NLP) techniques are typically used to extract the sentiment expressed in text, thereby eliminating the need for human intervention in sentiment

prediction [5]. The sentiment analysis techniques developed for a general domain may not be well-suited for the financial domain [6]. The words generally considered negative may not be negative in financial texts, such as the word “liability.” Loughran and McDonald [7] built a lexicon for financial text. The sentiment analysis using this lexicon counts the number of positive and negative words present in the sentence to assign the polarity, and it does not consider the semantic meaning of the text. A transformer model trained on financial text will be able to comprehend financial terminology [8]. These models can be fine-tuned with smaller datasets for sentiment analysis tasks. The FinBERT model [9], [10], a pre-trained model on financial text that is fine-tuned for sentiment analysis, has outperformed other sentiment analysis techniques on financial text.

Traditional machine learning models, such as Support Vector Machines (SVMs) and Random Forests, have been widely applied for stock price and return forecasting due to their

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ability to capture nonlinear relationships among financial variables [11], [12], [13], [14]. However, these models treat each observation independently or rely on limited historical lags, making them ineffective in capturing the temporal dependencies inherent in financial time series [15], [16], [17]. To overcome these limitations, deep learning (DL) architectures particularly long short-term memory (LSTM), gated recurrent unit (GRU), and simple recurrent neural networks (RNN) have been extensively employed for time-series forecasting [18], [19], [20], as they learn long-term dependencies and temporal correlations directly from data, providing a more accurate and dynamic representation of market behavior. The temporal Kolmogorov-Arnold network (TKAN), introduced in [21] and inspired by the Kolmogorov-Arnold Network (KAN) [22], is specifically designed for handling time series data. Given that stock prices follow a temporal structure, the TKAN model is well-suited for stock market prediction tasks.

Causal analysis aims to determine whether a causal relationship exists between two or more time series of data or variables [23]. Such relationships can be linear or non-linear, and both forms are possible between stock returns and sentiment. If sentiment features such as polarity and subjectivity exhibit a causal relationship with index returns, incorporating them can enhance the prediction of index return signs. Hence, identifying the cause of a model's performance improvement or decline becomes essential.

Most studies on Indian stock index forecasting rely on generic features such as price and volume, with limited exploration of financial news sentiment [24], [25], [26], [27], [28], [29]. Traditional machine learning models also struggle to capture temporal dependencies and long-term patterns, thereby limiting their predictive accuracy [30]. To address these gaps, this paper proposes a methodology for predicting next-day Nifty50 return signs by integrating domain-specific sentiment features with historical return data. Pre-trained transformer models extract polarity, reflecting market sentiment, and subjectivity, distinguishing factual content from opinions. Sequential dependencies are modeled using deep learning architectures, including LSTM, GRU, SimpleRNN, and TKAN. The influence of sentiment features is examined through causal analysis, while interpretability is ensured through SHapley Additive Explanations (SHAP) and a streamlined architecture. This approach highlights the novelty of combining domain-specific sentiment, temporal modeling, and causal analysis to improve short-term forecasting of Indian stock indices.

The key contributions of this work are:

- Compiled a dataset of financial news for Nifty50 companies through web scraping.
- Developed deep learning models to predict index return direction using historical return signs, polarity, and subjectivity.
- Compared the proposed methodology with existing approaches.

- Analyzed causal relationships between sentiment features and stock returns using Granger Causality and Transfer Entropy.
- Evaluated feature importance with SHAP to quantify the impact of polarity and subjectivity.
- Performed trading simulations to demonstrate practical effectiveness.

The remainder of the paper is organized as follows: Section II discusses the related works. The section III discusses the proposed methodology. The results and discussion are detailed in section IV, and the conclusion of the work is drawn in section V.

II. RELATED WORK

The literature frequently uses a variety of factors, including technical indicators, fundamental ratios, and historical stock data, to forecast stock movement and prices. It has been demonstrated that these characteristics improve prediction accuracy. However, stock movement can also be influenced by other market conditions. The Efficient Market Hypothesis [31] states that stock prices take into account all available information. Global events, news, and social media posts all affect stock performance.

The authors Chiong et al. [32] in their work interpreted the financial data using sentiment analysis. They used a sliding window mechanism to capture the historical pattern. The polarity and subjectivity of the financial text were extracted using TextBlob. A novel ensemble of SimpleRNN, GRU, and LSTM models is proposed, and it achieved the state-of-the-art result on the Adhoc Dataset.

Kim et al. [33] employed FinBERT to compute sentiment scores on financial news. The LSTM model was trained using stock price data and sentiment to forecast the S&P 500 index movement. Based on error metrics such as MSE, RMSE, and MAE, the predictive ability of LSTM models was evaluated with and without sentiment data. The empirical findings showed that the prediction accuracy for all three measures increased when sentiment scores were incorporated using the LSTM model.

Srivastava et al. [34] used historical stock data, tweets, and financial news to forecast the price movements of individual stocks of Nifty50 companies. They used TextBlob and SentimentAnalyzer for sentiment analysis. Feature selection is done using the SelectKBest algorithm. LSTM, random forest (RF), support vector machine (SVM), and gradient boosting regressor are employed for movement prediction. The results showed that the LSTM model performed well in predicting prices.

Dahal et al. [35] research employed gated GRU and LSTM to forecast the stock price of S&P 500 using stock features alone and combining stock features with financial news data. A comparative study was conducted to determine the importance of incorporating financial news into stock price prediction. Sentiment analysis of financial news was conducted using the VADER Sentiment analyzer.

The experiment's findings demonstrated that using financial news data improves forecast accuracy compared to relying on fundamental stock attributes.

Deng et al. [36] proposed an explainable eXtreme gradient boosting (XGBoost) method for predicting the stock index. The Shapley Additive exPlanations (SHAP) and the gain function of XGBoost are used to determine the importance of sentiment in the prediction. Their research findings demonstrated that the inclusion of sentiment improved model performance, and it also revealed that institutional investors' sentiments are more important than those of individual investors and foreign investors.

Recurrent neural networks (RNNs), long short-term memory networks (LSTMs), and convolutional neural networks (CNNs) are the most commonly used deep learning technologies in stock time series prediction. Gulmez [37] optimized a deep LSTM network with the artificial rabbits optimization (ARO) for stock price prediction. LSTM-ARO is compared to an artificial neural network model, three different LSTM models, and an LSTM model optimized using the genetic algorithm. These models are evaluated using MSE, MAE, MAPE, and R2 criteria. The results showed that LSTM-ARO outperformed the other models.

Htun et al. [38] constructed feature vectors from multiple previous relative returns and applied the random forest, SVM, and LSTM models to predict whether a stock can return 2% more than its index in the following ten days. The LSTM model outperformed the machine learning models.

Singh et al. [39] introduced a method for stock market prediction leveraging an ensemble model. They applied a wavelet-based denoising technique as a preprocessing step, followed by an ensemble approach that combined LSTM, CNN, and temporal convolutional networks (TCN) through meta-learning. Their research focused on predicting the Nifty50 index.

Temporal Kolmogorov-Arnold networks (TKANs) [21] is a new neural network architecture inspired by KAN and the LSTM. TKAN combines the strengths of both networks, composed of recurring Kolmogorov-Arnold networks (RKANs) Layers, which embed memory management. This advancement has the potential to facilitate multi-step time series forecasting with improved precision and effectiveness.

Since the majority of the work was done on US data, a more thorough examination of predictive models for index return sign prediction in the Indian context is required, taking into account the influence of news sentiment. Most studies have employed TextBlob and VADER models for sentiment analysis [32], [34], [35], [40]; however, models trained on financial sentiment tasks would be more suitable for accurate insights. The DL models, such as LSTM, GRU, and LSTM, are widely used for stock price and movement prediction. The TKAN model is a newly introduced model for time series data, and its performance can be compared with DL models.

III. METHODOLOGY

In this work, we predict the nominal return sign of the Nifty 50 index by incorporating the news sentiments. The Figure. 1 depicts the overall methodology. The methodology starts by collecting the historical data of the Nifty 50 index from Yahoo Finance¹ and news data corresponding to the companies listed in Nifty50 from the moneycontrol² website. The data is collected from January 2020 to May 2024. Then the news data is fed into FinBERT for extracting polarity and subjectivity using the mDeBERTa V3 model. We calculated the market return sign from the historical data for the entire period. We used a sliding window approach to capture the historical data. The data is fed into various deep-learning models, including SimpleRNN, GRU, LSTM, and TKAN, as well as an ensemble of SimpleRNN, TKAN, GRU, and LSTM, to predict the next-day return sign. The implemented models were evaluated using accuracy, balanced accuracy, precision, recall, F1-score, and AUC score metrics.

A. DATA COLLECTION

In the data collection phase, financial news and stock data were gathered from January 2020 to May 2024. The movement of the Nifty50 index is influenced by news about companies featured on the index. Therefore, we scraped financial news for Nifty 50-listed companies from the Moneycontrol website. The news dataset has 2,719 entries (statistics can be found in Table 1, each representing a news article. The dataset used in this study is publicly available at the GitHub repository.³ The columns included in the dataset are as follows:

- **Company** (String): A unique identifier or name for the company mentioned in the news article.
- **Date Published** (Datetime): The precise date and time the article was published on Moneycontrol.
- **Author** (Dictionary): Information about the article's author.
- **Headline** (String): The main title or headline of the news article.
- **Description** (String): A concise summary or abstract of the article's content.
- **Article Body** (String): The complete text content of the news article.
- **Tags** (List of Strings): Keywords or categories assigned to the article for easy organization and search.
- **URL** (String): The direct URL to the original news article on the Moneycontrol website.

The daily stock market data for the Nifty50 index is collected from the Yahoo Finance platform. The collected data are:

¹<https://finance.yahoo.com/>

²<https://www.moneycontrol.com/>

³https://github.com/ChaithraNLPPProjects/Nifty50_Financial_News.git

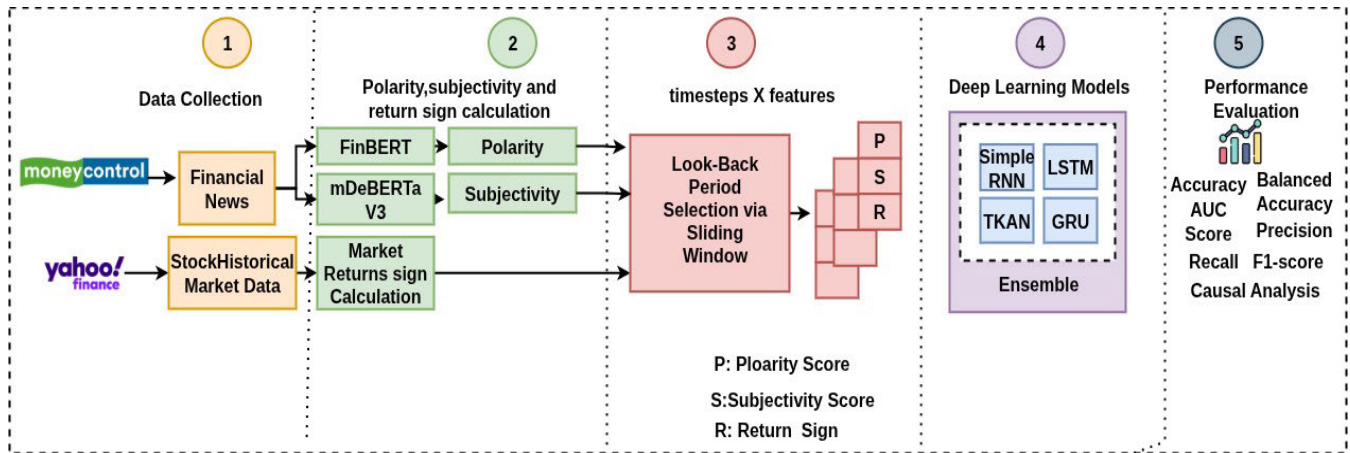


FIGURE 1. Proposed methodology for index return sign prediction.

TABLE 1. News dataset statistics.

Data Field	Unique Values
Total Entries	2,719
Companies	50
Authors	98
Headlines	2,522
Descriptions	2,642
Article Bodies	2,671
Tags	1,450
URLs	2,672

- **Open:** The price at which stock initially trades on a trading day.
- **High:** The maximum price a stock reaches during a trading day.
- **Low:** The minimum price a stock reaches during a trading day.
- **Close:** The price at which a stock closes during a trading day.

A sample structure of the stock data collected is shown in Table 2.

TABLE 2. Nifty50 index sample data.

Date	Open	High	Low	Close
2024-05-22	22576.60	22629.50	22483.15	22597.80
2024-05-23	22614.10	22993.60	22577.45	22967.65
2024-05-24	22930.75	23026.40	22908.00	22957.10
2024-05-27	23038.95	23110.80	22871.20	22932.45

B. SUBJECTIVITY, POLARITY AND RETURN SIGN CALCULATION

In this section, the extraction of polarity and subjectivity from financial news, as well as the calculation of index return signs, is explained in detail.

1) POLARITY

Polarity helps analyze whether the news has positive or negative sentiment, which can influence investor perceptions and market reactions. FinBERT [9] is a pre-trained transformer model fine-tuned for financial sentiment analysis. The FinBERT model has outperformed other sentiment analysis models on financial texts. Therefore, we employed FinBERT to determine the polarity score for financial news. The model and tokenizer used in our work were loaded from the repository on Hugging Face.⁴ The FinBERT model predicts labels as neutral, negative, or positive. The hyperbolic tangent (tanh) function was applied to the FinBERT model's logit output to translate labels into polarity scores between -1 and 1. A polarity value of -1 indicates a very negative sentiment, and a value of 1 indicates a very positive sentiment.

2) SUBJECTIVITY

Subjectivity helps identify whether the text presents opinions or facts, making it easier to assess the reliability of the information and its potential impact on financial decisions. We used the mDeBERTa V3 pre-trained transformer model, which is fine-tuned for subjectivity detection in news sentences. Subjectivity values range from 0 to 1, with 0 representing completely objective statements and 1 indicating highly subjective content. The mDeBERTa-v3 model and tokenizer are loaded from the repository⁵ on Hugging Face. The table 3 presents the calculated polarity and subjectivity scores for sample news articles.

3) STOCK MARKET RETURN CALCULATION AND SLIDING WINDOW APPROACH

The nominal return for a stock on a specific day t is calculated using the following equation 1:

$$\text{Nominal Return}_t = \frac{\text{Closing Price}_t - \text{Opening Price}_t}{\text{Opening Price}_t} \quad (1)$$

⁴<https://huggingface.co/ProsusAI/finbert>

⁵<https://huggingface.co/GroNLP/mdebertav3-subjectivity-english>

TABLE 3. Example of financial news sentiment with polarity and subjectivity scores.

Date	Symbol	News	Polarity	Subjectivity
2020-01-16	ADANIENT	CBI books Adani Enterprises, 3 ex-NCCF officials for alleged irregularities in coal supply contract	−0.9233	0.0564
2020-01-20	ADANIENT	Adani Enterprises denies wrongdoing amid allegations of coal supply contract irregularities	−0.9600	0.0654
2020-02-05	ADANIENT	Adani Enterprises net profit surges to Rs 383 cr in December quarter	0.9688	0.0461

where:

- Closing Price_{*t*} is the closing price of the stock on day *t*.
- Opening Price_{*t*} is the opening price of the stock on day *t*.

Nominal returns are then encoded as 1 for positive values, signaling an upward movement, and 0 for negative values, signaling a downward movement.

Using the sliding window method, we considered the previous five days' return signs, polarity, and subjectivity as features for the prediction. Five prior return signs were chosen for this study because the stock market operates on a five-day trading week.

C. DATASET FEATURES

The dataset included three input features: a stock return sign, polarity and subjectivity of the previous five time steps. Table 4 provides the statistical properties of the dataset used for the study. Since the financial news data was not available on every trading day, we considered neutral sentiment; i.e., for the polarity score, we set it to 0, and for subjectivity, we set it to 0.5, allowing the model to differentiate between the presence and absence of news data.

D. DEEP LEARNING MODELS

Deep learning models, particularly the recurrent neural network (RNNs) family, are efficient in handling sequential data [41], [42], [43]. Our study employed SimpleRNN, GRU, LSTM, and an ensemble of SimpleRNN, GRU, LSTM, and TKAN, optimized by particle swarm optimization (PSO), to forecast the sign of stock market returns.

The SimpleRNN model is designed to capture temporal patterns in sequential data [18], [44], [45]. It consists of a SimpleRNN layer with a tanh activation function, followed by a dropout layer for regularization. Finally, a fully connected layer with a sigmoid activation function is used to generate the output. The summary of the SimpleRNN architecture is given in Table 5.

The GRU model represents an advancement in capturing short to medium-range temporal sequences efficiently [46], [47]. GRU layer enhances memory retention and computational efficiency. The tanh activation function enables the model to retain critical information over extended sequences, which is crucial for predicting trends in dynamic financial markets. The model architecture summary is presented in Table 6.

The Long Short-Term Memory (LSTM) model is designed to capture long-term dependencies and temporal patterns in time-series data [48], [49], [50], [51]. LSTM mitigates the vanishing gradient problem through a memory cell with gated, additive state updates, allowing gradients to flow effectively across extended time steps. The architecture employed in this study is summarized in Table 7. To prevent overfitting, dropout regularization is applied within the LSTM layers, randomly deactivating units during training, which improves the model's generalization on unseen data. Finally, dense layers with sigmoid activation functions follow the LSTM layers, refining the extracted features and enhancing predictive accuracy.

The KAN architecture, as described by [22], is based on the Kolmogorov-Arnold representation theorem, which states that any continuous function with multiple variables can be expressed as a combination of univariate functions and addition operations. The KAN model utilizes B-spline curves to parameterize these univariate functions, offering improved performance and interpretability over traditional MLPs. The TKAN [21] architecture extends the KAN model to handle temporal sequences by integrating an LSTM cell. This combination allows the network to retain past information through memory management mechanisms, making it suitable for tasks involving complex sequential data. The major components of TKAN include:

- **RKAN Layers:** These layers manage short-term memory from previous states.
- **Gating Mechanism:** This helps manage information flow by deciding what to retain or discard.

The TKAN architecture consists of a single TKAN layer with 32 units that captures temporal patterns, followed by a dropout layer to improve generalization. A dense layer then reduces the dimensionality to a single unit for prediction. The complete architecture is presented in Table 8.

The ensemble of TKAN, SimpleRNN, GRU, and LSTM is also employed to enhance prediction accuracy. To optimize the ensemble model's predictive performance, particle swarm optimization (PSO) was employed to determine the optimal weights assigned to each model's predictions [30]. The PSO algorithm iteratively adjusted these weights to maximize classification accuracy on the validation set. The fitness function employed for PSO minimized the negative of the accuracy metric, ensuring that the weights were assigned to maximize predictive performance effectively. The final

TABLE 4. Statistical properties of the dataset used for the study.

Variables	Input/Output	Minimum	Maximum	Mean	Standard Deviation
Return Sign	Input	0	1	0.4965	0.5002
Polarity	Input	-0.9599	0.9705	0.1641	0.4565
Subjectivity	Input	0.0449	0.7046	0.2765	0.2026

TABLE 5. Summary of the SimpleRNN model architecture.

Layer (type)	Output Shape	Activation Function
SimpleRNN	(None, 1, 32)	tanh
Dropout	(None, 32)	
Dense	(None, 1)	sigmoid
Total params	1,121 (4.38 KB)	
Trainable params	1,121 (4.38 KB)	
Non-trainable params	0 (0.00 Byte)	

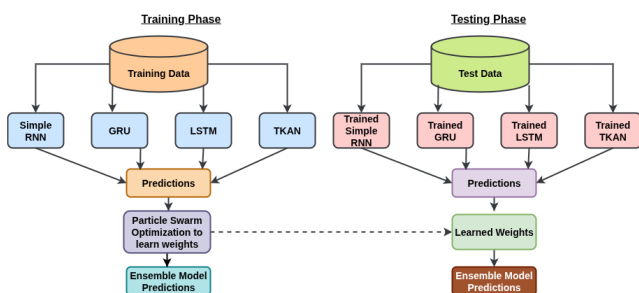
TABLE 6. Summary of the GRU model architecture.

Layer (type)	Output Shape	Activation Function
GRU	(None, 5, 64)	tanh
Dropout	(None, 5, 64)	
GRU	(None, 32)	tanh
Dropout	(None, 32)	
Dense	(None, 1)	sigmoid
Total params	22,305 (87.13 KB)	
Trainable params	22,305 (87.13 KB)	
Non-trainable params	0 (0.00 Byte)	

output, denoted as $f(x)$, generated by the model when presented with input x is determined by the ensemble methodology outlined in Equation 2.

$$w_1 \text{TKAN}(x) + w_2 \text{SimpleRNN}(x) + w_3 \text{GRU}(x) + w_4 \text{LSTM}(x) \\ w_1 + w_2 + w_3 + w_4 \quad (2)$$

where w_1, w_2, w_3 and w_4 are the weights assigned for model predictions. The Figure 2 illustrates the workflow of the ensemble modeling process.

**FIGURE 2.** Workflow of the ensemble model.

In all the models, the input is structured as a 5×3 matrix, where the dimension of 5 denotes the temporal sequence length, i.e., time steps, and the dimension of 3 denotes the number of features considered at each time step.

TABLE 7. Summary of the LSTM model architecture.

Layer (type)	Output Shape	Activation Function
LSTM	(None, 1, 32)	tanh
Dropout	(None, 32)	
Dense	(None, 1)	sigmoid
Total params	4,385 (17.13 KB)	
Trainable params	4,385 (17.13 KB)	
Non-trainable params	0 (0.00 Byte)	

TABLE 8. Summary of the TKAN model architecture.

Layer (type)	Output Shape	Activation Function
TKAN	(None, 1, 32)	<i>spline_order : 3, grid_size : 10</i>
Dropout	(None, 32)	
Dense	(None, 1)	sigmoid
Total params	3,383 (13.21 KB)	
Trainable params	3,383 (13.21 KB)	
Non-trainable params	3,383 (13.21 KB)	

E. PERFORMANCE EVALUATION

The implemented models were evaluated using accuracy, balanced accuracy, area under the curve, precision, recall, and F1 score, as shown in Equations 3 - 8.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (3)$$

$$\text{Balanced Accuracy} = \frac{1}{2} \left(\frac{\text{TP}}{\text{TP} + \text{FN}} + \frac{\text{TN}}{\text{TN} + \text{FP}} \right) \quad (4)$$

$$\text{AUC ROC} = \frac{1}{2} + \frac{\text{TP}}{2(\text{TP} + \text{FN})} - \frac{\text{FP}}{2(\text{FP} + \text{TN})} \quad (5)$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (6)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (7)$$

$$\text{F1 Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (8)$$

We conducted a causality test to examine the relationship between stock returns and polarity, as well as stock returns and subjectivity. We used the parametric Granger causality test and the non-parametric transfer entropy approach to conduct causal analysis. The Granger causality test, proposed by [52], is a statistical hypothesis test used to assess whether a time series can be used to predict another time series [53], [54]. The test is conducted to examine whether polarity and subjectivity features are important for return sign prediction. We used the *grangercausalitytests* function from

the *statsmodels* package in Python to perform the Granger causality test.

Transfer entropy measures and captures the pairwise causal direction between time series [55], [56], [57]. Stock returns and sentiment features, i.e., polarity and subjectivity, may exhibit both linear and non-linear relationships. Transfer entropy can capture both linear and nonlinear relationships, in contrast to Granger causality, which captures only linear relationships. We used the R package *RTransferEntropy* routines to quantify and test the information flow between the two time series measurements with Shannon and Rényi transfer entropy.

IV. RESULTS AND DISCUSSION

This section presents the results of the implemented models and baseline methods. The dataset is divided into an 80:20 split for training and testing, while preserving the temporal order to suit the time-series models. Each model is executed 30 times, and both the maximum and average accuracies from these runs are reported. To assess the predictive relevance of polarity and subjectivity, we conducted SHAP and causal analyses. Furthermore, an ablation study was performed to validate the contribution of individual features to model predictions.

- **Baseline Methods:** For comparison, we implemented baseline models, including XGBoost and random forest, with lag periods aligned to the timesteps used in our deep learning models. For both baselines, the number of estimators was set to 100, and the maximum tree depth was set to 3. In addition, we reproduced the best-performing model from [32], referred to as EnsembleRNN, for evaluation on our dataset. The EnsembleRNN model incorporates sentiment features extracted using TextBlob, along with five-day lagged returns and two sentiment features, polarity and subjectivity, as inputs. The architecture combines SimpleRNN, GRU, and LSTM layers, following the implementation details described in the original paper.

Table 9 reports the performance of all models across different temporal window sizes. Overall, the results indicate that a shorter window size ($t = 5$) yields the most effective predictive performance. At $t = 5$, TKAN achieves the highest accuracy of 0.5531, balanced accuracy of 0.5553, and AUC of 0.5408, outperforming GRU, LSTM, and SimpleRNN. Although LSTM records a slightly higher F1 score (0.5467), TKAN provides the most consistent performance across all metrics. As the window size increases to $t = 10$ and $t = 20$, the predictive performance of all models generally declines, suggesting that longer temporal dependencies may introduce noise or redundant information rather than improving learning efficiency. This trend suggests that a 5-day window effectively captures the most relevant short-term temporal patterns in stock movements; therefore, a 5-day window size is adopted for subsequent experiments.

Table 10 presents a comparative performance analysis of deep learning and baseline models using historical return sign, polarity, and subjectivity features. The results clearly show that the TKAN outperforms all baseline and recurrent models in both maximum and average settings. Specifically, TKAN achieves the highest accuracy of 0.5760, balanced accuracy of 0.5792, and AUC of 0.5446, indicating superior discriminative capability. Even under the average scenario, TKAN maintains strong and consistent results (accuracy = 0.5531), confirming its robustness across different experimental runs. Among the baseline models, random forest performs competitively in terms of recall (0.6053) and F1 score (0.5656), suggesting its strength in identifying positive return signals, although it falls short in overall balanced performance. SimpleRNN, GRU, and LSTM exhibit moderate performance, with LSTM (avg) reaching the highest F1 Score of 0.5467 among them but still lagging behind TKAN in overall accuracy and AUC. The ensemble model achieved an accuracy of 0.5112. This decline in performance suggests that the models included in the ensemble captured similar patterns, and their combination did not introduce sufficient diversity, thereby reducing the overall effectiveness of the ensemble.

We further examined the causal influence of news polarity and subjectivity on stock returns using the Granger causality test, and the results are summarized in Table 11. The analysis reveals that polarity exhibits a significant causal relationship with stock returns, particularly at higher lag orders (lags 3–5), where the p-values are below 0.01. This indicates that the polarity of news can help predict subsequent stock movements. In contrast, subjectivity shows no significant causal link with returns across all lag lengths, as reflected by p-values greater than 0.05. This suggests that the degree of subjectivity in news does not contain predictive information for stock returns. To validate these findings beyond linear dependencies, we further applied the transfer entropy test, a non-linear extension of the Granger causality framework.

The outcomes of the transfer entropy test are summarized in Table 12, offering a non-linear perspective on the causal interactions between sentiment features and stock returns. Compared to the linear Granger causality test, transfer entropy captures information flow in both directions, thus identifying asymmetric dependencies. The results reveal that while polarity to stock returns shows a weak causal effect (TE = 0.0134, $p = 0.2167$), it is not statistically significant. However, the reverse direction of stock returns to polarity is statistically significant ($p = 0.0267$), indicating that market movements may influence the polarity of subsequent news coverage rather than the other way around. In contrast, subjectivity demonstrates no significant causal interaction with stock returns in either direction ($p > 0.1$). This finding is consistent with the Granger causality results, reinforcing that subjectivity does not contribute meaningfully to predicting stock index movements.

Overall, the comparative analysis between the Granger and transfer entropy results suggests that polarity exhibits limited

TABLE 9. Performance comparison of models across different timesteps.

Time Steps	Model	Accuracy	Balanced Accuracy	AUC	Precision	Recall	F1 Score
t = 5	TKAN	0.5531	0.5553	0.5408	0.5942	0.5123	0.5363
	SimpleRNN	0.5048	0.5045	0.5030	0.5300	0.5091	0.5187
	GRU	0.5402	0.5426	0.5271	0.5732	0.4956	0.5299
	LSTM	0.5453	0.5465	0.5257	0.5746	0.5228	0.5467
	Ensemble	0.5112	0.5121	0.5147	0.5387	0.4942	0.5147
t = 10	TKAN	0.5510	0.5519	0.5378	0.5772	0.5310	0.5456
	SimpleRNN	0.4744	0.4746	0.4702	0.4951	0.4708	0.4820
	GRU	0.5357	0.5381	0.5293	0.5645	0.4792	0.5161
	LSTM	0.5530	0.5545	0.5390	0.5894	0.4735	0.5220
	Ensemble	0.4975	0.4990	0.4968	0.5216	0.4637	0.4901
t = 20	TKAN	0.5250	0.5250	0.5119	0.5531	0.5246	0.5342
	SimpleRNN	0.4905	0.4899	0.4893	0.5166	0.5021	0.5082
	GRU	0.5362	0.5396	0.5202	0.5722	0.4736	0.5166
	LSTM	0.5104	0.5001	0.5119	0.5257	0.6991	0.5951
	Ensemble	0.4983	0.4958	0.4955	0.5234	0.5426	0.5303

TABLE 10. Performance comparison of deep learning models using historical return sign, polarity, and subjectivity. (Best results for maximum and average accuracy are highlighted.)

Category	Model	Accuracy	Balanced Accuracy	AUC	Precision	Recall	F1 Score
Proposed	TKAN (Max)	0.5760	0.5792	0.5446	0.6146	0.5175	0.5619
	TKAN (Avg)	0.5531	0.5553	0.5408	0.5942	0.5123	0.5363
	SimpleRNN (Max)	0.5392	0.5385	0.5199	0.5625	0.5526	0.5575
	SimpleRNN (Avg)	0.5048	0.5045	0.5030	0.5300	0.5091	0.5187
	GRU (Max)	0.5530	0.5596	0.5319	0.6049	0.4298	0.5026
	GRU (Avg)	0.5402	0.5426	0.5271	0.5732	0.4956	0.5299
	LSTM (Max)	0.5668	0.5713	0.5266	0.6111	0.4825	0.5392
	LSTM (Avg)	0.5453	0.5465	0.5257	0.5746	0.5228	0.5467
	Ensemble (Max)	0.5622	0.5655	0.5296	0.6000	0.5000	0.5455
	Ensemble (Avg)	0.5112	0.5121	0.5147	0.5387	0.4942	0.5147
Baseline	XGBoost (Avg)	0.4700	0.4699	0.4557	0.4954	0.4737	0.4843
	Random Forest (Max)	0.5115	0.5065	0.4764	0.5308	0.6053	0.5656
	Random Forest (Avg)	0.4854	0.4799	0.4674	0.5088	0.5880	0.5451
	Ensemble RNN (Max)	0.5459	0.5540	0.5014	0.6056	0.3772	0.4649
	Ensemble RNN (Avg)	0.5011	0.5041	0.4952	0.5292	0.4377	0.4782

TABLE 11. Granger causality results.

Variables under test	No of Lags	F Value	p Value
Stock returns and polarity	1	3.0521	0.0811
	2	1.6505	0.1928
	3	5.3950	0.0011
	4	4.0330	0.0031
	5	3.0576	0.0098
Stock returns and subjectivity	1	3.1402	0.0769
	2	1.8511	0.1579
	3	1.2705	0.2835
	4	0.8941	0.4670
	5	0.7759	0.5673

predictive power in a non-linear context, while subjectivity remains statistically insignificant in influencing or being influenced by stock returns.

The low causal link between polarity and stock return may be because of the following reasons:

- The available trading data spanned 1090 trading days, equivalent to four years of data, while the news data was only available for 628 days. For the causal analysis techniques to properly comprehend how one event or factor influences another over time, data over a long period is necessary [58].
- News might not significantly impact the Nifty50 index movement because the Nifty50 index is less volatile compared to the movements of individual stocks.

We also performed SHAP analysis on the model predictions to determine the relative importance of each feature. We took a subset from the test data to perform SHAP analysis.

The global feature importance in Figure 3 indicates that polarity is the strongest contributor to the model's predictive performance, followed by historical returns, whereas subjectivity has the least impact. When we examine the temporal decomposition shown in 4, the results reveal that recent information is more valuable than older signals. In particular,

TABLE 12. Transfer entropy results.

X and Y	Direction	TE	Eff. TE	Std. Err.	p-value
Polarity and Stock return	$X \rightarrow Y$	0.0134	0.0025	0.0045	0.2167
	$Y \rightarrow X$	0.0202	0.0088	0.0047	0.0267
Subjectivity and Stock return	$X \rightarrow Y$	0.0166	0.0069	0.0048	0.1400
	$Y \rightarrow X$	0.0113	0.0011	0.0045	0.3000

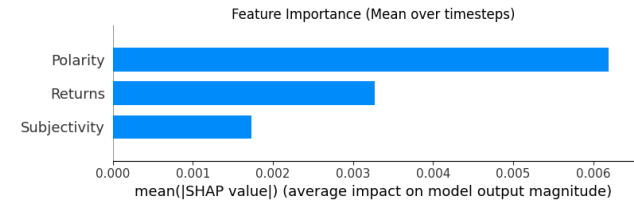


FIGURE 3. Global feature importance plot.

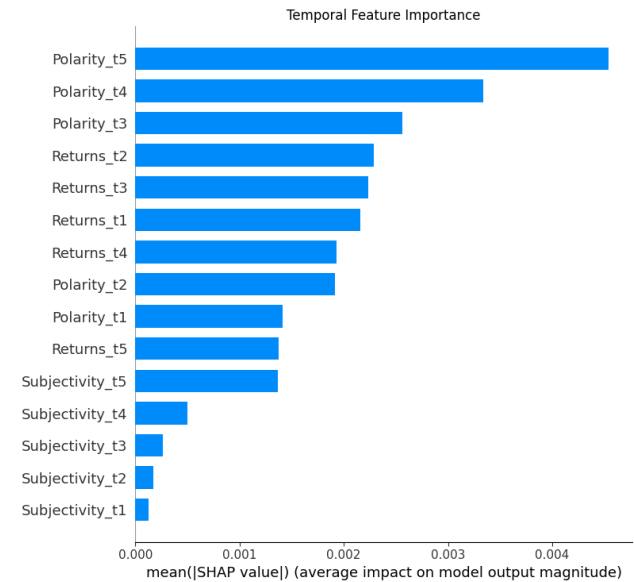


FIGURE 4. Temporal feature importance plot.

Polarity_{t5} (the most recent lag) has the strongest impact among all features, while earlier lags of polarity and subjectivity exhibit progressively lower importance. The SHAP waterfall plot Figure 5 illustrates how features contribute to a specific prediction: recent polarity tends to push the output upward, whereas other lags such as Returns_{t1} or Returns_{t2} may offset this by pulling it downward, with subjectivity and polarity having smaller but non-negligible effects. Collectively, these findings highlight that the model relies heavily on recent return dynamics and sentiment-related features, with subjectivity emerging as the most influential input category overall.

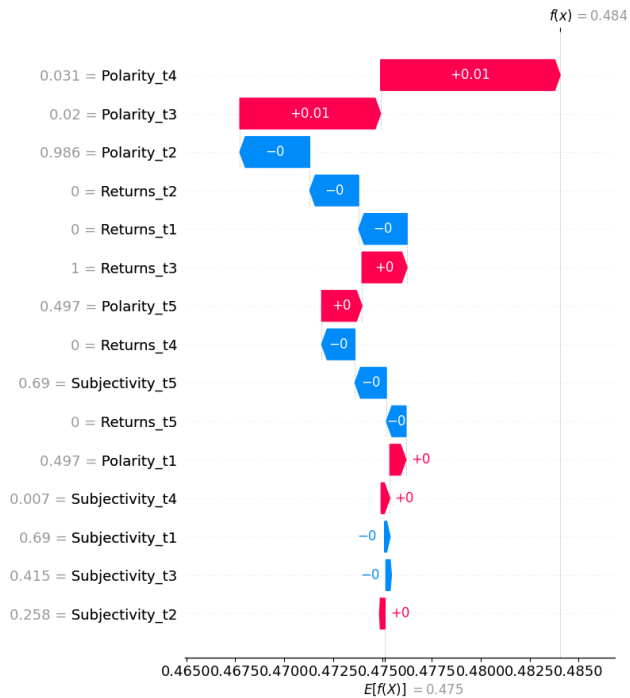


FIGURE 5. SHAP waterfall plot illustrating the contribution of lagged features.

A. ABLATION STUDY

Following the causal and SHAP analyses, we conducted an additional study to assess the importance of individual features in prediction using deep learning models. The objective was to examine whether the contribution of sentiment features aligns with the insights obtained from the SHAP and causal analyses. We evaluated the following model configurations:

- Model with historical return sign feature only
- Model with historical return sign and polarity
- Model with historical return sign and subjectivity

The ablation results presented in Tables 13, 14, and 15 reveal how the polarity and subjectivity of sentiment components affect model performance when combined with historical return signs.

When trained only with historical return signs, the TKAN model demonstrated superior performance with a maximum accuracy of 0.5760 and an average accuracy of 0.5573,

TABLE 13. Performance comparison of DL models using historical return signs only. (The best results for maximum and average accuracy are highlighted.)

Model	Accuracy	Balanced Accuracy	AUC	Precision	Recall	F1 Score
TKAN (Max)	0.5760	0.5684	0.5786	0.5775	0.7193	0.6406
TKAN (Avg)	0.5573	0.5587	0.5795	0.5895	0.5310	0.5542
SimpleRNN (Max)	0.5207	0.5214	0.5487	0.5472	0.5088	0.5273
SimpleRNN (Avg)	0.4969	0.4995	0.5381	0.5244	0.4488	0.4823
GRU (Max)	0.5622	0.5641	0.5471	0.5941	0.5263	0.5581
GRU (Avg)	0.5507	0.5529	0.5532	0.5833	0.5088	0.5431
LSTM (Max)	0.5530	0.5568	0.5416	0.5914	0.4825	0.5314
LSTM (Avg)	0.5482	0.5497	0.5447	0.5776	0.5213	0.5479
Ensemble (Max)	0.5668	0.5676	0.5553	0.5943	0.5526	0.5727
Ensemble (Avg)	0.5243	0.5260	0.5471	0.5524	0.4927	0.5199

TABLE 14. Performance comparison of DL models using historical return sign and polarity. (The best results for maximum and average accuracy are highlighted.)

Model	Accuracy	Balanced Accuracy	AUC	Precision	Recall	F1 Score
TKAN (Max)	0.5760	0.5684	0.5779	0.5775	0.7193	0.6406
TKAN (Avg)	0.5555	0.5580	0.5786	0.5943	0.5085	0.5417
SimpleRNN (Max)	0.5622	0.5660	0.5517	0.6022	0.4912	0.5411
SimpleRNN (Avg)	0.5296	0.5323	0.5480	0.5611	0.4807	0.5168
GRU (Max)	0.5622	0.5637	0.5630	0.5922	0.5351	0.5622
GRU (Avg)	0.5427	0.5477	0.5648	0.5846	0.4491	0.5071
LSTM (Max)	0.5853	0.5884	0.5647	0.6250	0.5263	0.5714
LSTM (Avg)	0.5628	0.5659	0.5556	0.5994	0.5061	0.5483
Ensemble (Max)	0.5714	0.5771	0.5688	0.6235	0.4649	0.5327
Ensemble (Avg)	0.5416	0.5452	0.5627	0.5780	0.4754	0.5212

TABLE 15. Performance comparison of DL models using historical return sign and subjectivity. (The best results for maximum and average accuracy are highlighted.)

Model	Accuracy	Balanced Accuracy	AUC	Precision	Recall	F1 Score
TKAN (Max)	0.5991	0.5983	0.5392	0.6195	0.6140	0.6167
TKAN (Avg)	0.5495	0.5502	0.5340	0.5812	0.5365	0.5470
SimpleRNN (Max)	0.5069	0.5049	0.5072	0.5299	0.5439	0.5368
SimpleRNN (Avg)	0.4805	0.4797	0.4925	0.5057	0.4953	0.4998
GRU (Max)	0.5853	0.5851	0.5446	0.6091	0.5877	0.5982
GRU (Avg)	0.5562	0.5548	0.5420	0.5781	0.5825	0.5782
LSTM (Max)	0.5806	0.5821	0.5338	0.6117	0.5526	0.5806
LSTM (Avg)	0.5482	0.5467	0.5282	0.5704	0.5760	0.5727
Ensemble (Max)	0.5760	0.5731	0.5374	0.5902	0.6316	0.6102
Ensemble (Avg)	0.5227	0.5212	0.5159	0.5451	0.5520	0.5479

outperforming other deep learning models. This indicates TKAN's ability to effectively capture short-term temporal dependencies from price-based features. The Ensemble model also performed competitively, achieving a maximum accuracy of 0.5668 and a precision of 0.5943, which suggests that combining multiple models helps stabilize predictions, although it lags slightly behind TKAN in average accuracy.

With the inclusion of polarity alongside return signs, the LSTM model demonstrated a significant improvement, achieving the highest average accuracy of 0.5628 and a maximum accuracy of 0.5853. This highlights LSTM's strength in leveraging sentiment polarity to capture investor mood shifts that influence short-term stock movement. The results also demonstrate that polarity contributes more predictive

information than subjectivity, as all models recorded marginal gains in both accuracy and AUC compared to the baseline setup.

When subjectivity was introduced instead of polarity, the GRU model emerged as the best performer, achieving a maximum accuracy of 0.5991 and an average accuracy of 0.5562. Interestingly, TKAN achieved the same highest maximum accuracy of 0.5991 in this configuration, suggesting that while subjectivity offers some predictive signal, it introduces more noise than polarity, leading to mixed improvements across models. The SimpleRNN model's performance dropped considerably in this setup, reaffirming its limitations in handling higher-level sentiment variations.

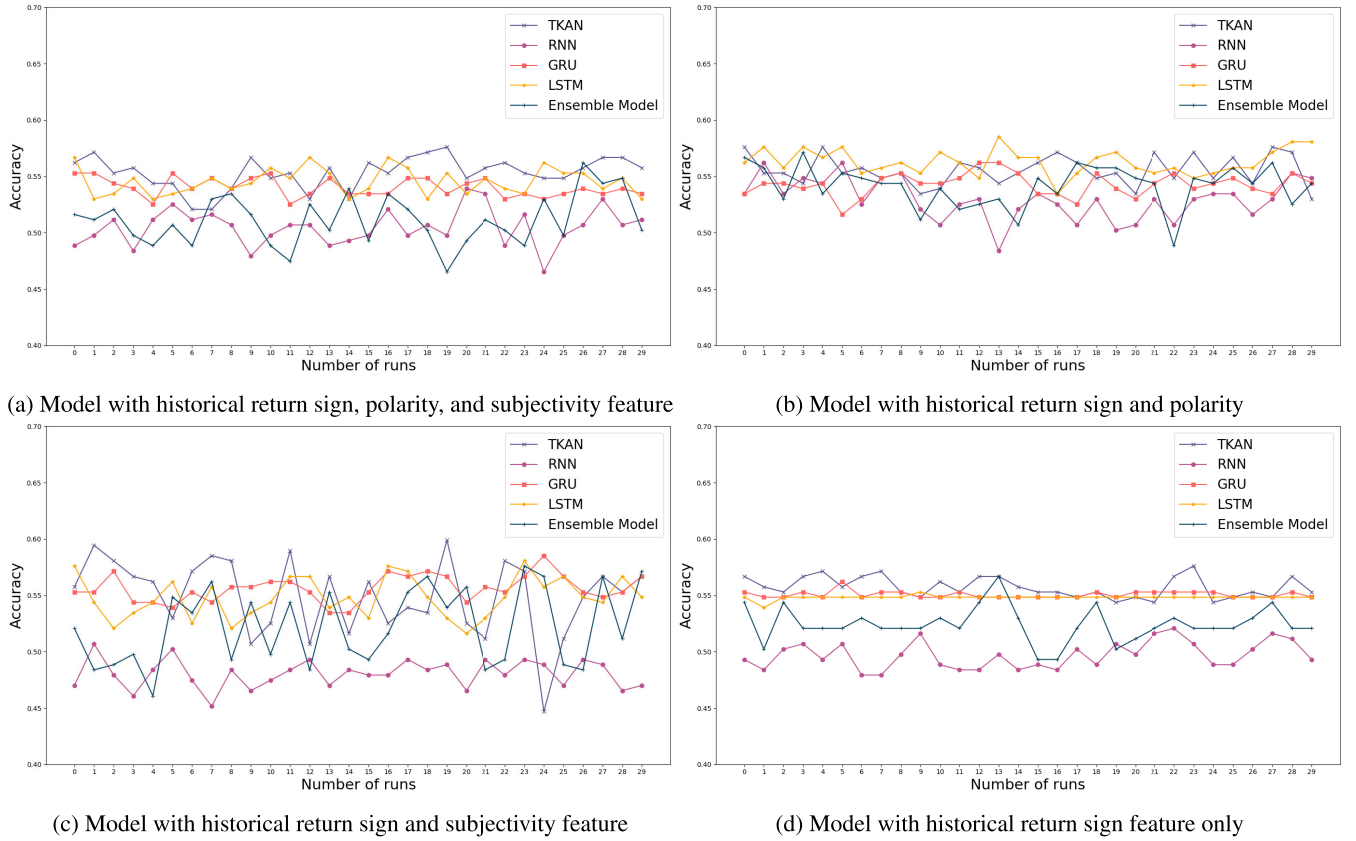


FIGURE 6. Comparison of performance metrics of deep learning models based on 30 experimental runs.

Overall, the ablation study suggests that integrating sentiment polarity yields the most consistent improvement in predictive performance across deep learning models, while subjectivity contributes selectively. Among the models, LSTM and TKAN consistently outperform others, demonstrating robust temporal and contextual learning capabilities.

Figure 6 presents the comparison results over 30 runs in terms of accuracy. The results indicate that when both return sign and polarity features are included, the models exhibit more stable performance with lower fluctuations compared to other feature combinations. This highlights polarity as a significant contributor to improving model predictions.

B. TRADING SIMULATION

To demonstrate the practical applicability of the sentiment-based index movement prediction models, we designed a trading algorithm that integrates model-generated signals to make buy and sell decisions. The complete trading procedure is summarized in Algorithm 1.

Although the index itself is not directly tradable, index-based ETFs exhibit similar performance to the underlying index. Hence, the simulation results can be effectively mapped to any corresponding index-based ETF. The algorithm takes as input daily stock prices $P_t = \{Open_t, Close_t\}$, model predictions $\hat{y}_t^m \in \{0, 1\}$ from each

deep learning model, initial capital C_0 , and a transaction cost rate γ . To mitigate the high transaction costs associated with daily trading, the model predictions and prices are resampled into biweekly intervals, with the first and last prices of each period used as the opening and closing values, respectively. A transaction cost of 0.05% per trade is applied to realistically account for market frictions.

During each biweekly period, if a model signals a buy ($\hat{y}_t = 1$) and no position is currently held, the algorithm invests the available capital minus transaction costs. Conversely, if a sell signal ($\hat{y}_t = 0$) is triggered and a position exists, the algorithm liquidates the holdings, accounting for transaction costs. The portfolio value V_t is updated at the end of each period, and standard performance metrics, such as total return, annualized return, and the Sharpe ratio, are calculated to assess the model's effectiveness.

$$\text{Total Return} = \left(\frac{V_T}{C_0} \right) - 1 \quad (9)$$

where V_T is the final portfolio value and C_0 is the initial capital invested.

$$\text{Annualized Return} = (1 + \text{Total Return})^{\frac{1}{T}} - 1 \quad (10)$$

where T is the duration of the investment in years.

$$\text{Sharpe Ratio} = \frac{\mu - r_f}{\sigma} \quad (11)$$

TABLE 16. Trading simulation results (results above baseline are highlighted in bold).

Category	Model	Total Return	Annual Return	Sharpe Ratio
Returns Only	TKAN	0.1146	0.1308	1.0096
	SimpleRNN	0.0126	0.0144	-0.6813
	GRU	0.1905	0.2186	1.6146
	LSTM	0.1619	0.1854	1.3203
	Ensemble	0.0503	0.0572	-0.0501
Returns with Polarity	TKAN	0.1893	0.2172	1.5667
	SimpleRNN	0.1795	0.2058	1.4958
	GRU	0.1840	0.2110	1.5276
	LSTM	0.1607	0.1841	1.3027
	Ensemble	0.1795	0.2058	1.4958
Returns with Subjectivity	TKAN	0.1130	0.1290	0.6516
	SimpleRNN	0.1660	0.1902	1.1822
	GRU	0.1282	0.1466	0.9479
	LSTM	0.1634	0.1871	1.3199
	Ensemble	0.0662	0.0754	0.1937
Returns with Subjectivity and Polarity	TKAN	0.2043	0.2346	1.5679
	SimpleRNN	0.0094	0.0107	-0.9103
	GRU	0.1572	0.1800	1.2355
	LSTM	0.1060	0.1209	0.6638
	Ensemble	0.1216	0.1389	0.7523
Baseline	BuyandHold	0.1678	0.1922	1.1002

where:

- μ : Mean return per period
- σ : Standard deviation of return per period
- r_f : Annual risk-free rate

Table 16 presents the trading simulation results under four feature scenarios: returns only, returns with polarity, returns with subjectivity, and returns with both polarity and subjectivity, compared to a Buy-and-Hold baseline.

When using returns only, the GRU model achieved the highest total return of 19.05% and a Sharpe ratio of 1.6146, slightly outperforming LSTM and TKAN, while the Ensemble and SimpleRNN models underperformed relative to the baseline.

Incorporating polarity significantly improved performance for most models. TKAN, GRU, SimpleRNN, and Ensemble all recorded higher total returns and Sharpe ratios than the Buy-and-Hold strategy, indicating that sentiment polarity provides valuable predictive information for trading decisions.

When subjectivity was included instead of polarity, improvements were more selective: SimpleRNN and LSTM showed modest gains, but TKAN and GRU exhibited lower Sharpe ratios, suggesting that subjectivity alone may introduce noise and reduce risk-adjusted performance.

The combination of polarity and subjectivity as features resulted in the best overall performance for TKAN, with a total return of 20.43%, an annualized return of 23.46%, and a Sharpe ratio of 1.5679, clearly outperforming both the baseline and all other models. Other models, such as GRU and LSTM, achieved moderate gains, whereas SimpleRNN and Ensemble performed poorly in this configuration.

TABLE 17. Statistical test results based on the Wilcoxon rank-sum test for accuracy evaluation (p-values less than 0.05 are highlighted in bold).

	SimpleRNN	GRU	LSTM	Ensemble Model
TKAN	1.16E-10	9.10E-05	1.63E-02	6.26E-09
SimpleRNN		3.82E-10	2.10E-10	3.62E-01
GRU			1.27E-01	3.00E-07
LSTM				1.02E-07

TABLE 18. Statistical test results based on the Wilcoxon rank-sum test for balanced accuracy evaluation (p-values less than 0.05 are highlighted in bold).

	SimpleRNN	GRU	LSTM	Ensemble Model
TKAN	4.48E-11	1.73E-04	9.25E-03	2.43E-09
SimpleRNN		2.14E-10	1.86E-10	2.46E-01
GRU			3.52E-01	4.27E-07
LSTM				1.41E-07

Overall, the trading simulation highlights that integrating sentiment-driven features, especially polarity and subjectivity, with historical returns significantly enhances the practical utility of deep learning models for real-world trading. TKAN consistently generated the most robust and profitable trading signals, demonstrating its ability to effectively capture market dynamics.

C. STATISTICAL ANALYSIS

We used the Wilcoxon rank-sum test to determine the statistical significance of the obtained results, as the implemented models produced varying accuracy across multiple runs. We used a two-sided p-value to check the hypothesis.

- Null Hypothesis: Models are significantly the same.
- Alternate Hypothesis: Models are significantly different.

Algorithm 1 Trading Strategy With Model Predictions**Input:**

Daily stock prices $P_t = \{Open_t, Close_t\}$ for $t = 1, \dots, T$

Model predictions $\hat{y}_t^m \in \{0, 1\}$, where $m \in \{\text{RNN, GRU, LSTM, TKAN, Ensemble}\}$

Initial capital C_0 , transaction cost rate γ

Output: Return, Sharpe Ratio

```

1: Resample  $P_t$  into biweekly prices:
    $Open_{bw} =$  first price of period,  $Close_{bw} =$ 
   last price of period
2: for each strategy  $s \in \{\text{models, Buy\&Hold}\}$  do
3:   Initialize  $cash \leftarrow C_0$ ,  $position \leftarrow 0$ ,  $V_0 \leftarrow C_0$ 
4:   for each biweekly period  $t = 1, \dots, T$  do
5:      $Signal_t \leftarrow$  prediction from strategy  $s$ 
6:     if  $Signal_t = 1$  and  $position = 0$  then
7:        $trade\_cost \leftarrow cash \cdot cost\gamma$ 
8:        $trade\_money \leftarrow cash - trade\_cost$ 
9:        $position \leftarrow trade\_money / Close_{bw,t}$ 
10:       $cash \leftarrow 0$ 
11:     else if  $Signal_t = 0$  and  $position > 0$  then
12:        $trade\_money \leftarrow position \cdot Close_{bw,t}$ 
13:        $cost \leftarrow trade\_money \cdot cost\gamma$ 
14:        $cash \leftarrow trade\_money - cost$ 
15:        $position \leftarrow 0$ 
16:     end if
17:      $V_t \leftarrow cash + position \cdot Close_{bw,t}$ 
18:   end for
19: end for
20: Compute metrics:
   Total Return  $= V_T / C_0 - 1$ 
   Annualized Return  $= (1 + \text{Total Return})^{1/Y} - 1$ 
   Sharpe Ratio  $= \frac{\mu - r_f}{\sigma}$ 
21: return Portfolio value series  $V_t$  and performance metrics

```

The p-values are below the threshold for TKAN, indicating that the models are statistically different when compared pairwise with respect to accuracy and balanced accuracy, as shown in Tables 17 and 18. The p-values are above the threshold for the pair [GRU, LSTM] and [SimpleRNN, Ensemble], suggesting that the models are not significantly different.

V. CONCLUSION

This study presents a novel approach for predicting the sign of stock index returns by integrating news sentiment with historical price data. The proposed deep learning models leverage polarity and subjectivity scores extracted using a domain-specific pretrained sentiment model, combined with returns incorporated via a sliding window approach. This integration allows the model to capture both temporal dependencies and sentiment-driven market signals, demonstrating a clear advancement over baseline strategies.

Our experimental results on the Nifty50 index highlight several key findings. First, incorporating polarity scores significantly improves predictive performance across all architectures, yielding a 5.2% improvement in average accuracy over the baseline. Subjectivity, in contrast, has a limited influence on model predictions, as confirmed by causality analyses, including the Granger test and transfer entropy. SHAP analysis further emphasizes that polarity is the primary driver of model decisions. Trading simulations demonstrate that models incorporating polarity, particularly TKAN, consistently outperform the Buy-and-Hold strategy, highlighting their practical applicability for short-term and day traders. Statistical tests indicate that TKAN performs significantly differently from other models, whereas LSTM and GRU models are statistically comparable.

Despite these promising results, the study has some limitations. The ensemble model suffers from reduced effectiveness due to the inclusion of multiple models that capture similar patterns, which limits the diversity gains. Moreover, the analysis relies on a single data source for news sentiment and does not account for broader macroeconomic variables, which could enhance predictive power.

Future research should focus on addressing these technical gaps by exploring more sophisticated ensemble strategies that maximize model diversity, integrating multi-source news and macroeconomic indicators, and applying advanced preprocessing techniques such as signal decomposition to denoise price movements. Additionally, real-world implementation considerations, including transaction costs and liquidity constraints, should be further incorporated into predictive and trading simulations to increase practical relevance.

Overall, the study demonstrates that sentiment-informed deep learning models, particularly TKAN, can provide actionable insights for stock index forecasting. The approach can aid traders and investment analysts in anticipating short-term market movements, optimizing trading strategies, and making informed decisions based on the combined analysis of historical returns and sentiment-driven signals.

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